

Panel Data II: Difference-in-Differences

Francisco Villamil

Applied Quantitative Methods II
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Today's goals

- Understand Difference-in-Differences as an extension of TWFE

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- Learn the 2×2 DiD setup and the parallel trends assumption

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- Use event studies to visualize dynamic effects and test pre-trends
- Understand the staggered DiD problem and modern solutions

Roadmap

From Fixed Effects to DiD

The DiD Estimator

Parallel Trends and Its Threats

Event Studies

Staggered DiD and Recent Advances

Wrap-up

Recap: what fixed effects does

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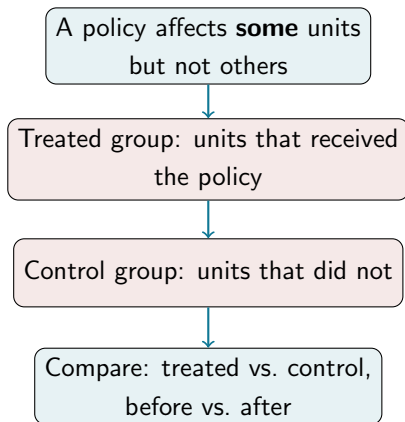
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- $\hat{\beta}$: identified from within-unit variation, net of common trends
- But TWFE treats x_{it} as a continuous variable that varies continuously over time
- What if the variation comes from a **specific, discrete intervention**?

A new question: policy interventions



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 - **Treatment:** fast-food restaurants in New Jersey (NJ)
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 - **Before:** Feb–Mar 1992; **After:** Nov–Dec 1992
- Pennsylvania did not change its minimum wage

The 2×2 DiD table

	Before	After
Control (PA)	$\bar{y}_{C,pre}$	$\bar{y}_{C,post}$
Treated (NJ)	$\bar{y}_{T,pre}$	$\bar{y}_{T,post}$

$$\hat{\delta}_{DiD} = (\bar{y}_{T,post} - \bar{y}_{T,pre}) - (\bar{y}_{C,post} - \bar{y}_{C,pre})$$

- Control group change: what would have happened **without** treatment

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- Control group change: what would have happened **without** treatment
- DiD subtracts this “counterfactual trend” from the treated group change

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The DiD formula

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- $\hat{\delta} = 0.59 - (-2.17) = +2.76$
- Interpretation: the minimum wage **raised** NJ employment by ≈ 2.76 FTEs relative to PA

The regression formulation

$$y_{it} = \alpha + \beta_1 \underbrace{Post_t}_{\text{time}} + \beta_2 \underbrace{Treat_i}_{\text{group}} + \delta \underbrace{(Post_t \times Treat_i)}_{\text{interaction = DiD}} + \varepsilon_{it}$$

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- β_1 : time trend (common to both groups)
- β_2 : pre-period difference between groups

Regression: recovering the 2×2 cells

	Before ($Post = 0$)	After ($Post = 1$)
Control ($Treat = 0$)	α	$\alpha + \beta_1$
Treated ($Treat = 1$)	$\alpha + \beta_2$	$\alpha + \beta_1 + \beta_2 + \delta$

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- The regression recovers $\hat{\delta}$ automatically via OLS

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- Always cluster standard errors at the **treatment level**

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The key assumption: parallel trends

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In the absence of treatment, the treated and control groups would have followed the **same time trend**.

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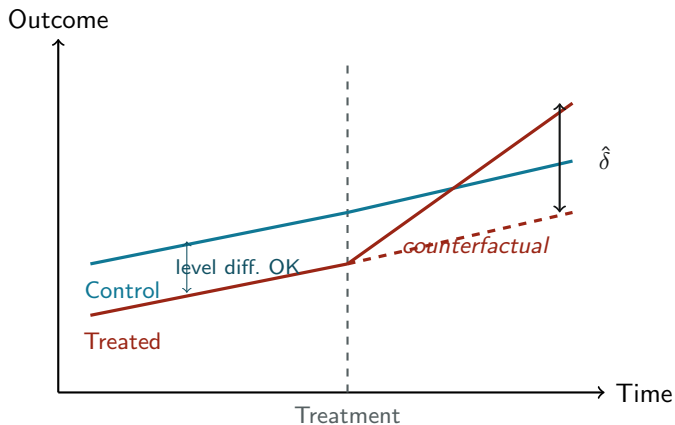
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- What this does **not** require:
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 - Pre-period outcomes can differ — just not the trends
- This is an **assumption**, not a testable fact
(we cannot observe the counterfactual)

Parallel trends: visualization



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- **Anticipation effects**

- Treated units change behavior before the official treatment date
- Example: firms start hiring/firing when the wage increase is announced

The parallel trends assumption says treatment would not have changed the outcome trajectory.

When is this plausible?

What makes NJ and PA a good comparison?

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 - **Dynamic effects**: does the effect grow, fade, or remain constant?
- This is the **event study** design

Event study regression

$$y_{it} = \alpha_i + \gamma_t + \sum_{k \neq -1} \delta_k \cdot \mathbf{1}[t - T_i^* = k] + \varepsilon_{it}$$

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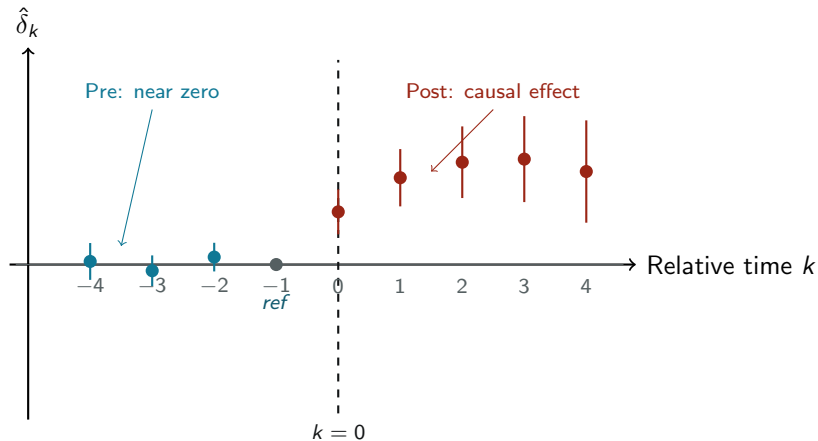
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- Unit and time FE absorb levels; δ_k captures relative changes

Event study: coefficient plot



Event study in R

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- `fixest` makes event studies very easy to run and plot

Reading an event study plot

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- **Symmetry:** sometimes useful to check behavior far pre-treatment

Roadmap

From Fixed Effects to DiD

The DiD Estimator

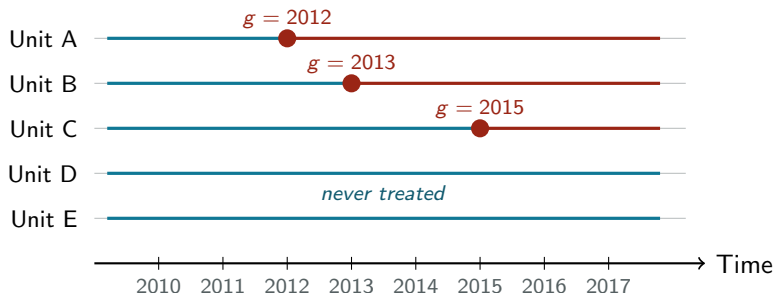
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Staggered treatment adoption



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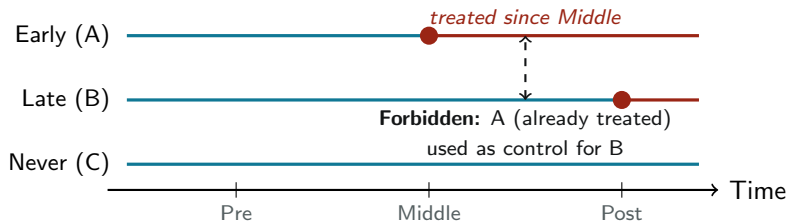
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- Result: TWFE can produce a **weighted average with negative weights**
 - Estimate may not correspond to any valid ATT

Negative weights: the intuition



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- Works even with **heterogeneous treatment effects**

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- In R: `etwfe` package by Grant McDermott

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emfx(m) # marginal effects = ATT
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Naive TWFE vs. modern estimators

	Naive TWFE	Callaway-Sant'Anna	ETWFE
Clean control groups	No	Yes	Yes
Heterogeneous effects	Biased	Handles	Handles
Negative weights	Possible	None	None
Pre-trend test	Via event study	Built-in	Via emfx
Implementation	feols	did	etwfe

- With simultaneous treatment (all treated at same time): TWFE is fine

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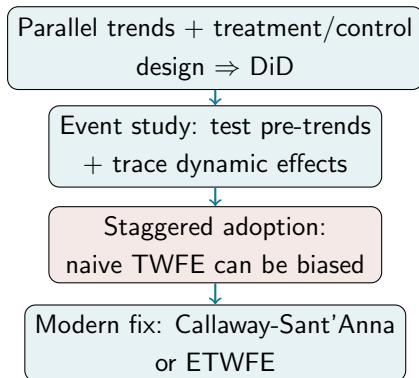
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DiD: putting it all together



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- Always **cluster SEs** at the treatment-assignment level

For next session

- Complete Assignment 6 (DiD application)
- Read the assigned paper (DiD design)
- Next session: Spatial Data (I)
 - Spatial data structures and visualization
 - Spatial autocorrelation
 - Spatial regression models

Questions?